

1 Norms and Regularizers

In MAP, we are looking for weights that maximizes $\log p(\mathcal{D}|\mathbf{w}) + \log p(\mathbf{w})$. Since we have explored plenty about the log likelihood term $\log p(\mathcal{D}|\mathbf{w})$, this question shall offer some love to the log prior term $\log p(\mathbf{w})$. Recall that when we assume $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \beta^{-1}\mathbf{I})$, we obtained $\log p(\mathbf{w}) = \text{const} - (\beta/2) \cdot \mathbf{w}^\top \mathbf{w}$, which (ignoring the constants) is exactly the L2 term that we add to the loss function. A natural question hence arises: what other regularizers can we obtain if we simply choose a different prior? Throughout this question, we shall assume $\mathbf{w} \in \mathbb{R}^M$ is M -dimensional.

Problem 1. Let $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\boldsymbol{\mu}, \beta^{-1}\mathbf{I})$ be multivariate Gaussian distribution with nonzero mean $\boldsymbol{\mu} \neq \mathbf{0}$ and covariance $\beta^{-1}\mathbf{I}$. By simplifying $\log p(\mathbf{w})$, work out what regularizer does this corresponds to.

Problem 2. For each dimension $m \in [M]$, let $p(w_m)$ be the Laplace distribution with mean 0 and scale parameter $b > 0$. As such, the joint density is

$$p(w) = \prod_{m=1}^M \frac{1}{2b} \exp\left(-\frac{|w_m|}{b}\right).$$

By simplifying $\log p(w)$, work out what regularizer does this corresponds to.

Problem 3. For each dimension $m \in [M]$, let $p(w_m) = \mathcal{U}(w_m|0, 1)$ be Uniform distribution with support $[0, 1]$ and let each $w_1, \dots, w_m$ be mutually independent. By observing the form of $\log p(\mathbf{w})$ or otherwise, interpret what effect does this prior has on your learning process.

2 Connection between Naive Bayes and Logistic Regression

Note: This tutorial sheet has been updated since it was first posted to add in the negative sign in the exponent of the logistic sigmoid. It was initially missing. We have updated the solutions accordingly. Some of you may have seen an alternative solution presented by your TAs (that is correct when the negative sign is missing). Here, we are streamlining the solution to be consistent with the lectures since the logistic sigmoid is defined as $\sigma(z) = \frac{1}{1+\exp(-z)}$.

We have learnt two approaches to classification: Logistic regression (discriminative) and naive Bayes (generative). What are the similarities and differences between the two approaches? If you are given a classification problem, which approach should you choose? We will explore these questions in this problem.

Here, we will have labels $t \in \{0, 1\}$. Recall that binary logistic regression model can be expressed in the following “probabilistic linear” form:

$$p(t = 1|\mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x})}$$

We want to show that Naive Bayes under the following assumptions results in a model with the above form:

1. Features are independent.
2. Features are Gaussian.
3. The variances are shared across classes.

Problem 1. Let us apply Bayes Rule to obtain $p(t = 1|x)$ from the generative term $p(\mathbf{x}|t = 1)$:

$$p(t = 1|\mathbf{x}) = \frac{p(\mathbf{x}|t = 1)p(t = 1)}{\sum_t p(\mathbf{x}, t)}.$$

Show that we can simplify this to the expression

$$p(t = 1|\mathbf{x}) = \frac{1}{1 + \exp\left(\log \frac{p(t=0)}{p(t=1)} + \log \frac{p(\mathbf{x}|t=0)}{p(\mathbf{x}|t=1)}\right)}. \quad (1)$$

Problem 2. Let $\mathbf{x} \in \mathbb{R}^M$. We assume the setup in naive Bayes where

$$p(\mathbf{x}|t = j) = \prod_{m=1}^M \mathcal{N}(x^m | \mu_j^m, v^m),$$

that is, the features of $\mathbf{x}$ are i.i.d. The two classes have distinct means $\{\mu_0^m\}_{m=1}^M$ and $\{\mu_1^m\}_{m=1}^M$ but shared variances $\{v^m\}_{m=1}^M$. For the prior over t , we assume a standard Bernoulli distribution $p(t) = \pi^t(1 - \pi)^{1-t}$. Let us try to simplify the denominator in the expression of Equation 1. The first term is

$$\log \frac{p(t = 0)}{p(t = 1)} = \log \frac{1 - \pi}{\pi}.$$

The second term requires more work, so let us break it down into steps. First, show that

$$\log \frac{p(\mathbf{x}|t=0)}{p(\mathbf{x}|t=1)} = \sum_{m=1}^M \log \frac{\exp\left(-\frac{(x^m - \mu_0^m)^2}{2v^m}\right)}{\exp\left(-\frac{(x^m - \mu_1^m)^2}{2v^m}\right)}.$$

Hint: remember that

$$\mathcal{N}(x^m | \mu_j^m, v^m) = \frac{1}{\sqrt{2\pi v^m}} \exp\left(-\frac{(x^m - \mu_j^m)^2}{2v^m}\right).$$

Problem 3. Now, it's a matter of algebra in simplifying the above expression. Show that you can arrive at the expression

$$\log \frac{p(\mathbf{x}|t=0)}{p(\mathbf{x}|t=1)} = \sum_m \left(\frac{\mu_0^m - \mu_1^m}{v^m} x^m + \frac{(\mu_1^m)^2 - (\mu_0^m)^2}{2v^m} \right).$$

Problem 4. Recall from Problem 1 that for logistic regression $p(t=1|\mathbf{x}) = (1 + \exp(-\mathbf{w}^\top \mathbf{x}))^{-1}$, where we absorb the bias term into $\mathbf{w}^\top \mathbf{x}$ using $\mathbf{w} = [w^0, w^1, \dots, w^M]^\top$ and $\mathbf{x} = [1, x^1, \dots, x^M]^\top$. Let us write the bias term explicitly as

$$p(t=1|x) = \frac{1}{1 + \exp(-w^0 - \sum_{m=1}^M w^m x^m)}.$$

By substituting the expressions for $\log(p(t=0)/p(t=1))$ and $\log(p(x|t=0)/p(x|t=1))$ that we have just derived into Equation 1, show that the weights from logistic regression represent the terms

$$\begin{aligned} w^0 &= -\log \frac{1-\pi}{\pi} - \sum_m \frac{(\mu_1^m)^2 - (\mu_0^m)^2}{2v^m} \\ w^m &= -\frac{\mu_0^m - \mu_1^m}{v^m}. \end{aligned} \tag{2}$$

In other words, logistic regression and naive Bayes with Gaussian features share the same form. The weights learnt in logistic regression are related to the Gaussian parameters learnt in naive Bayes as given by Equation 2.

Bonus. We have answered how logistic and naive Bayes are similar. But how are they different for a given classification problem? This paper answers this question. It turns out that logistic regression performs better than naive Bayes with more training samples as it reaches a lower asymptotic error. However, naive Bayes approaches its higher asymptotic error with fewer samples. Practically speaking if you have a small dataset, you may consider naive Bayes, while if you have a large dataset you should adopt logistic regression.